Semiparametric Value-at-Risk and Expected Shortfall with a Real-Time Misspecification Score

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Abstract

Value-at-Risk and Expected Shortfall are typically built from a parametric distribution that is hard to beat on most days. A real-time score from trailing GARCH-standardized residuals predicts, for each asset and date, when a nonparametric alternative is more accurate; the advantage concentrates in the score's top decile (+3.0% pinball, DM 10.5) and is negligible elsewhere. A jump-robust refit attributes most of it to variance overstatement after a shock, with a smaller but significant part surviving the refit. An amortized estimator attains the lowest joint (VaR, ES) loss against standard benchmarks and extends to newly listed assets without refitting.

Keywords: Value-at-Risk; Expected Shortfall; misspecification; quantile regression; extreme value theory; amortized estimation.

JEL classification: C14; C22; C52; C58; G17; G28.

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Introduction

A risk model is judged on its worst days, and those are the days on which a fixed parametric tail is most likely to be wrong. The error is costly in both directions. A forecast that runs high ties up capital against risk that never arrives, and one that runs low leaves too little against the loss that does. Deciding when a more flexible model is worth its added complexity is therefore a practical question. It is not only a question for the bank trading desks that Basel regulates. Asset managers, hedge funds, and any function that forecasts a conditional tail face the same choice. Value-at-Risk (VaR) and Expected Shortfall (ES), the forecasts at issue, summarize the downside risk of a trading book: the lower-tail quantile of the conditional return distribution, and the average loss beyond it. Three model classes produce these forecasts, and they differ in how much of the conditional return distribution they fix by assumption (Kuester et al., 2006; Nieto and Ruiz, 2016). Fully parametric filters, GARCH and its leverage and skew extensions (Bollerslev, 1986; Glosten et al., 1993), deliver conditional scale dynamics that are hard to improve within the ARCH family (Hansen and Lunde, 2005) but read the tail off an assumed innovation density. Semiparametric filters keep that scale and estimate the innovation law from the standardized residuals, empirically (Barone-Adesi et al., 1999), through an extreme-value tail (McNeil and Frey, 2000), or by modeling the quantile directly (Engle and Manganelli, 2004), and Francq and Zakoïan (2015) supply the two-step estimator's asymptotic theory. Distribution-free conditional quantile estimation by pinball-loss minimization is itself an econometric idea (Koenker and Bassett, 1978), and machine learning has scaled it up. Its members here are gradient-boosted quantile trees, neural implicit quantile networks (IQN) (Dabney et al., 2018), and the quantile-network generator of generative Bayesian computation (GBC) (Polson and Sokolov, 2023). GBC is a simulation-based posterior method. It becomes a conditional quantile learner once its forward simulator is replaced by observed state and return pairs, as Section 1 does. These learners place no distributional restriction on the innovation law, but they must learn it from data and from a chosen state vector. Machine-learning models now forecast realized volatility better than heterogeneous-autoregressive benchmarks (Bucci, 2020; Christensen et al., 2023), and when pooled across a cross-section of US stocks they transfer to names outside the training set (Zhang et al., 2024). This paper asks whether a comparable advantage exists for the quantile learners studied here, whether it reaches the regulatory tail, and on which days it appears.

Industry practice sits between these classes. Under the Basel Fundamental Review of the Trading Book (FRTB), the workhorse internal models are historical simulation (HS), by far the most common method in bank disclosures (Pérignon and Smith, 2010), and its GARCH-filtered variant (FHS) (Barone-Adesi et al., 1999). The regulatory choice is not between a fixed model and a proprietary one but between a prescribed standardized formula and a bank's own internal model, and many banks have moved back toward the standardized approach as internal models became costly to maintain. The same classical toolkit is what the buy side runs: asset managers and hedge funds, though outside Basel, forecast VaR and ES with historical simulation, parametric filters, and Monte Carlo, largely through vendor risk systems. The comparison in this paper is therefore against the methods in standard use across the industry, not a regulatory strawman. Forecasts are judged by VaR exception counts at 99% and 97.5%, and the reported measure is Expected Shortfall at the 97.5% level. That measure is more than a diagnostic. The reported ES sets the market-risk capital a bank holds against its book (Basel Committee on Banking Supervision, 2019), so errors are costly in both directions. A forecast that runs high ties up capital that could fund positions, while one that runs low raises the multiplier on the charge as exceptions accumulate. A method that claims to beat the industry standard must therefore beat HS and FHS on a jointly consistent $(\mathrm{VaR}, \mathrm{ES})$ score at the 97.5% level (Fissler and Ziegel, 2016), since ES alone cannot be scored (Gneiting, 2011), and it must do so under exception-test discipline (Nolde and Ziegel, 2017). Beating a vanilla GARCH on average loss is accordingly not enough. Neither is it trivial, because bank internal models have historically failed to beat a simple GARCH fitted to their own profit and loss (Berkowitz and O'Brien, 2002). The GARCH filter is the floor of this paper's comparisons rather than its target.

The advantage of a flexible model over this parametric benchmark is not spread evenly across days, and that unevenness is the paper's subject. On the roughly ninety percent of asset-days (one name on one trading date) when residual-shape stress is low, the estimator we use and the parametric benchmark are statistically indistinguishable on forecast accuracy, so holding the flexible model carries no detectable penalty (Section 4 reports the intervals). The gains are concentrated in the minority of days when residual-shape stress is high. Locating that minority in advance, from an observable signal, is the paper's central task. It is a conditional predictive-ability question in the sense of Giacomini and White (2006): does a signal known at the forecast date predict which of two forecasts will be more accurate on that date? We construct such a signal, a real-time misspecification score, and show that it orders the flexible model's advantage. We then build a risk model that exploits the ordering while matching the benchmark elsewhere. Because its residual-shape model is pooled across names and conditioned on characteristics, it also delivers forecasts for assets that have no return history at all, something no per-asset model, parametric or not, can do. Pooled estimation is known to improve volatility forecasts out of sample (Bollerslev et al., 2018; Zhang et al., 2024); here it is carried to the day-one limit and into the tail.

The score is built from the oldest specification diagnostics for a GARCH filter, the excess kurtosis and asymmetry of its standardized residuals (Bollerslev, 1987; Hansen, 1994). It is computed on a trailing window so that it is available in real time, in the spirit of the density-forecast diagnostics of Berkowitz (2001). What is new is the use to which these diagnostics are put: computed on recent residuals, they predict when distribution-free quantile methods improve on parametric VaR/ES. Where residual-shape stress is high the flexible estimator has a large and significant advantage, and where it is low the advantage is statistically indistinguishable from zero in the design panel and a small positive in the point-in-time universe, with no detectable loss in either. The score's content is an ordering of magnitudes. The edge concentrates in the top decile, and there is no sharp threshold below which it vanishes.

This paper makes three contributions. First, to the evaluation of VaR and ES forecasts and the comparison of competing conditional models, it introduces the misspecification frontier: a real-time, per-asset, per-day score that predicts when distribution-free quantiles beat a GARCH-t benchmark, with significance from Diebold–Mariano tests on date-averaged loss differentials (the per-date DM of Section 3). The frontier is validated across four universes, namely CRSP US equities (top-decile edge +2.98%, DM 10.5), foreign exchange (hyperinflation tier, pooled DM 4.12), twenty-six country indices (a developed-to-frontier-market gradient), and forty-three cross-asset instruments. A central contribution is to identify what that advantage is made of. Measured against a jump-robust GARCH that cannot overshoot its variance after a large shock, we show that most of it is the standard filter's post-outlier scale error, with a smaller but significant component surviving a robust scale (the top-decile +2.98% becomes +0.3–0.5%, DM 2.5–4.7). This attribution, drawn on a 200-name panel and tied to a live score, is new to the flexible-versus-parametric tail-risk literature. The Korea-2026 crash, which unfolded under circuit breakers and left residual kurtosis in its ordinary range, provides a falsification case that sharpens the claim: a price crash is not residual misspecification, and GARCH-t wins there, as the frontier predicts.

Second, to global pooled forecasting and cross-asset transfer, it shows that a single shape model pooled across hundreds of names (amortized, in the machine-learning sense that one fit serves every asset) carries transferable information about conditional tail shape: it replaces per-asset estimation of the innovation shape and transfers to names never seen in training. In its characteristics-only form it forecasts VaR and ES for assets with no return history at all, the day-one regime in which no per-asset model exists.

Third, to semiparametric conditional tail-risk forecasting, it contributes the estimator that captures the frontier. The design follows the two-step logic of filtered historical simulation and GARCH-EVT (Barone-Adesi et al., 1999; McNeil and Frey, 2000). It keeps the parametric filter's conditional scale, the component the ARCH family already estimates well once a leverage term is allowed (Hansen and Lunde, 2005), and replaces only the innovation law where it fails. The forecast is therefore a GARCH conditional scale times a state-conditioned nonparametric residual quantile, with an extreme-value left tail and an optional split-conformal overlay. The resulting estimator is co-best with CAViaR (Engle and Manganelli, 2004) in the 90% Model Confidence Set (Hansen et al., 2011) and improves significantly on GARCH-t, FHS, EWMA, HS, and GJR-skew-t. Its EVT-tailed accuracy layer attains the lowest joint $(\mathrm{VaR}, \mathrm{ES})$ score, the zero-homogeneous FZ0 loss of Fissler and Ziegel (2016) and Patton et al. (2019), against the daily benchmarks in standard use, GARCH-t and FHS, at both FRTB levels. That ranking survives annual refitting, with exception counts near nominal through a 2008-crisis window. It also keeps the lowest FZ0 against the FZ-estimated dynamic ES models of Patton et al. (2019) and Taylor (2019), each estimated per name by FZ0 minimization, winning at 1% and tying Taylor at 2.5%. Two results motivate the score without supplying its mechanism. The first is the standard identity that excess pinball loss is locally quadratic in the quantile wedge, scaled by the local density (Gneiting, 2011; Ehm et al., 2016; Steinwart and Christmann, 2011), so that a wedge at a level generates regret there. The second, a tail-wedge proposition, shows how a misspecified innovation shape opens such a wedge in the tail even when the scale is right (McNeil and Frey, 2000; Francq and Zakoïan, 2015). The mechanism itself is what the decomposition identifies: the score flags mostly scale error, and the shape wedge is the smaller part that survives.

The loss profile is asymmetric in the score. When the frontier is quiet the estimator ties or edges the parametric benchmark (calm-quintile edge ≈ 0). In the top decile the improvement is +2.98%, and on the hyperinflation and capital-control currencies, where the parametric assumption breaks entirely, it is 12–14%. These are reductions in a consistent scoring function for the quantile (Gneiting, 2011), not a monetary welfare measure. A percentage pinball edge is a percentage reduction in average quantile loss at the quoted level. Because the ranking of misspecified forecasts can depend on which consistent scoring function is used (Patton, 2020), the joint FZ0 score of Section 5 serves both as the regulatory comparison and as a check on that ordering under a second score.

The paper proceeds as follows. Section 1 places the method against the industry benchmark and the forecasting literature. Section 2 develops the semiparametric estimator and the two results that motivate the frontier. Section 3 sets out the data and the forecast-evaluation protocol. Section 4 presents the misspecification frontier, and Section 5 reports the joint VaR and ES evaluation. Section 6 covers cross-sectional transfer, longer horizons, and cross-asset evidence, and Section 7 concludes.

1 Related literature and benchmark models

1.1 Risk measures and the FRTB discipline

For a return r_{t+1} with conditional law F_t, the level-α return-space quantile is $q_t^\alpha = F_t^{-1}(\alpha)$ (negative in the left tail). The return-space Expected Shortfall is $e_t^\alpha = \mathbb{E}[r_{t+1} \mid r_{t+1} \leq q_t^\alpha]$ for continuous F_t. In general $e_t^\alpha = \alpha^{-1} \int_0^\alpha F_t^{-1}(u)\, du$, the coherent form used throughout (Artzner et al., 1999; Acerbi and Tasche, 2002). The regulatory VaR and ES are their negatives (the sign convention is fixed once in Section 2.1). FRTB (Basel Committee on Banking Supervision, 2019) sets the capital metric at a stressed $\mathrm{ES}_{97.5}$ over a liquidity-adjusted horizon (ten days for the base bucket), while backtesting remains VaR-exception based. More than twelve exceptions at 99%, or thirty at 97.5%, in a 250-day year pushes a trading desk off the internal-models approach. The empirical bar for "beating the industry standard" is therefore joint. It requires accuracy on the joint (VaR, ES) score at 97.5% (Fissler and Ziegel, 2016; Patton et al., 2019), since ES alone cannot be scored (Gneiting, 2011), and on pinball, together with unconditional (Kupiec, 1995) and independence (Christoffersen, 1998) exception tests at both levels. Significance is assessed by per-date Diebold–Mariano tests (Diebold and Mariano, 1995) and the Model Confidence Set (Hansen et al., 2011). Real-time monitoring of VaR and ES forecast adequacy is developed by Hoga and Demetrescu (2023); the score of Section 2.6 is a cross-sectional, per-asset monitor of the same forecasts, used to rank states rather than to signal a structural break.

1.2 The parametric family, the simulation family, and CAViaR

The comparison set spans the models banks run. It includes historical simulation, EWMA with the RiskMetrics decay $\lambda = 0.94$ (J.P. Morgan/Reuters, 1996) (the long-memory successor in Zumbach, 2007 is not run here), GARCH(1,1)-t (Bollerslev, 1986, 1987), and GJR-GARCH-skew-t (Glosten et al., 1993; Hansen, 1994), which adds a leverage term and a skewed innovation density. It also includes GARCH-FHS (Barone-Adesi et al., 1999; Hull and White, 1998), which pairs a parametric scale with empirical standardized residuals and is the filtered method that the comparison literature ranks closest to the top (Kuester et al., 2006; Nieto and Ruiz, 2016). The strongest non-nested semiparametric rival in the set is SAV-CAViaR (Engle and Manganelli, 2004), to which the comparison literature is less favorable at the 1% level (Kuester et al., 2006). The SAV (symmetric absolute value) specification models the quantile directly by regression quantiles (Koenker and Bassett, 1978), $q_t(\tau) = \beta_0 + \beta_1\, q_{t-1}(\tau) + \beta_2\, |r_{t-1}|$, the quantile analogue of an absolute-value GARCH recursion (the indirect-GARCH CAViaR is the analogue of the squared recursion). The comparison set does not include GARCH-EVT (McNeil and Frey, 2000), which Kuester et al. (2006) rank first among classical methods. Its filter-then-tail logic is instead absorbed into the estimator's own EVT stage (Section 2), so the EVT-tailed layer is its analogue within the comparison set.

1.3 Distribution-free conditional quantiles and GBC

For the pooled, cross-sectional design the closest antecedent is Zhang et al. (2024), who forecast realized volatility with machine-learning models pooled across stocks and document transfer to previously unseen names. That transfer is evidence of a shared volatility mechanism, which our amortized design also exploits. For the VaR comparison the antecedent is Kuester et al. (2006), and for machine-learning volatility forecasting Bucci (2020) and Christensen et al. (2023). Deep neural-network quantile regression applied directly to VaR is the recent comparison closest to our nonparametric stage (Chronopoulos et al., 2024). The differences delimit this paper's contribution: their object is average realized-variance forecast performance, whereas ours is state dependence in conditional tail-risk forecast performance, with the full return distribution and its regulatory tail held to calibration, exception-test, and capital discipline. The question here is where a flexible model helps at all: on roughly ninety percent of asset-days it does not improve on the parametric benchmark, which cautions against extrapolating average machine-learning superiority to the tails. Our amortization is pushed to the cold-start limit (day-one forecasts from characteristics alone) and stress-tested against own-history benchmarks at each listing age. On tabular state, gradient-boosted trees outperform the implicit quantile network, the reverse of the intraday ranking in Zhang et al. (2024), where neural networks dominate tree-based models. At their daily horizon that dominance already disappears, and our target is a residual quantile curve rather than realized variance, with a different feature set.

The nonparametric family estimates $Q_\theta(\tau \mid x_t)$ directly by pinball-loss empirical risk minimization (ERM) (Koenker and Bassett, 1978; Steinwart and Christmann, 2011). Its two members here are gradient-boosted quantile trees (GBM) (Friedman, 2001; Meinshausen, 2006) and the implicit quantile network of Dabney et al. (2018), the architecture at the core of GBC (Polson and Sokolov, 2023; Nareklishvili et al., 2025). In the IQN the quantile level τ is itself fed into the network, so one set of weights outputs the whole quantile curve rather than one grid point at a time. Read as ERM on observed pairs, GBC's quantile generator coincides with a conditional quantile network trained on data (Dabney et al., 2018; Koenker and Bassett, 1978): the forward simulator is not needed, and the same object applies to observational time series (Section 2.5). The GBM is the shape model used throughout, and the IQN is retained as a benchmark. Its generative reading (draw $\tau \sim U(0,1)$, output $Q_\theta(\tau \mid x)$) is a capability we note without using, since sampling and posterior uncertainty are outside this paper's scope. Generalized Bayes supplies the frame for this estimator without being a component of it. Pinball ERM is the flat-prior mode, and the large-ω limit, of the Gibbs posterior $\pi_n(\theta) \propto \exp\{-\omega \sum_t \ell_\theta(z_t)\}\pi(\theta)$, which is built from a loss in place of a likelihood and targets the risk minimizer directly (Jiang and Tanner, 2008; Bissiri et al., 2016). We use the posterior itself only as a hierarchical shrinkage check in Section 2.3. Distribution-free finite-sample coverage comes from split-conformal prediction (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019), which shifts a predicted quantile by an order statistic of held-out calibration errors so that marginal coverage holds under exchangeability with no distributional model. Adaptive conformal inference (Gibbs and Candès, 2021) replaces the fixed shift by an online update whose long-run coverage holds under arbitrary distribution shift, at the cost of the finite-sample guarantee. Barber et al. (2023) bound the coverage loss when exchangeability fails, the case for time-ordered residuals.

2 Semiparametric VaR and ES model

Each estimator below can be reimplemented from the formulas and algorithm boxes given. A glossary of acronyms is in the Online Appendix.

2.1 Notation and objects

Let r_t be the return of an asset on day t and $\mathcal{F}_{t-1}$ the information available before t. The conditional τ-quantile of r_t given $\mathcal{F}_{t-1}$ is $Q_t(\tau) = \inf\{q : \Pr(r_t \le q \mid \mathcal{F}_{t-1}) \ge \tau\}$, the number the return falls below with probability τ. One sign convention is used throughout. The paper works in *return space*, with $q_\tau = Q_t(\tau) < 0$ in the left tail and $e_\tau = \tau^{-1} \int_0^\tau Q_t(u)\, du \le q_\tau$, the coherent form of Expected Shortfall (Artzner et al., 1999; Acerbi and Tasche, 2002). This is the convention of the FZ0 scoring function $(v, e < 0)$ (Fissler and Ziegel, 2016; Patton et al., 2019) and of each table, where ES values such as -6.37 are return-space conditional means. The regulatory *loss-space* numbers are their negatives, $\mathrm{VaR}_\tau^{\mathrm{loss}} = -q_\tau$ and $\mathrm{ES}_\tau^{\mathrm{loss}} = -e_\tau$, used only when quoting FRTB levels. The two are never mixed in one expression.

The *standardized residual* $z_t = (r_t - \mu_t)/\sigma_t$ is the return with the model's conditional mean and scale divided out. If the parametric model is right, the z_t are i.i.d. draws from one fixed law. The state vector s_t collects trailing realized volatility, the misspecification scores of Section 2.6, and lags, and is what the nonparametric quantiles condition on. A quantile level $\tau \sim \lambda$ is drawn from a base distribution λ. The learned generator $H_\phi(s, \tau)$ maps (state, level) to a quantile in the sense of Polson and Sokolov (2023): $z \overset{D}{=} H(s, \tau)$ with $\tau \sim U(0, 1)$ is the inverse-CDF (von Neumann) representation of sampling.

Loss and metric. All quantile models are trained and scored by the pinball (check) loss

$$\rho_\tau(u) \;=\; u\,(\tau - \mathbb{I}\{u < 0\}), \qquad u = z - q, \tag{1}$$

whose expectation is minimized by a conditional τ-quantile (Koenker and Bassett, 1978; Gneiting, 2011)—uniquely when that quantile is unique—the way squared error elicits the mean. Averaging (1) over $\tau \in (0, 1)$ targets the whole distribution at once (the integrated loss is the continuous ranked probability score up to a factor of two, as recorded below). The loss is density-free, so each estimator below dispenses with a likelihood.

Choice of scoring function. We score by pinball, a consistent scoring function for the quantile (Gneiting, 2011), at the decision-relevant levels rather than by a global proper scoring rule, for three reasons. (a) The continuous ranked probability score (CRPS), the proper scoring rule for the whole distribution (Gneiting and Raftery, 2007), is exactly the integral of the pinball loss over all levels, $\mathrm{CRPS}(F, y) = 2\int_0^1 \rho_\tau(y - F^{-1}(\tau))\, d\tau$ (Gneiting and Ranjan, 2011). Scoring by CRPS therefore weights quantile levels uniformly, so a 1% tail occupies 1% of the integral. The quantile-weighted CRPS of Gneiting and Ranjan (2011) reweights the levels for this reason, and scoring at fixed levels is its point-mass case. Downside risk lives at fixed regulatory levels ($\tau \in \{0.01, 0.025\}$), and the models in this paper differ almost entirely in the tail, and CRPS would average those differences away against a body where all competitors agree. For the same reason, training uses tail-weighted τ-sampling. (b) Expected Shortfall is not elicitable on its own (Gneiting, 2011) (no loss function is minimized by the true ES alone), but the pair $(\mathrm{VaR}, \mathrm{ES})$ is, under the joint scoring functions of Fissler and Ziegel (2016), whose zero-homogeneous member, the FZ0 loss of Patton et al. (2019), is the one Section 5 uses. The FZ-scored re-evaluation of the comparison set is an extension specified in advance (Section 5). (c) Likelihood is unavailable by design: several competitors have no tractable density, and likelihood scores would silently exclude them.

2.2 The final model: the residual-hybrid estimator

Figure 1 summarizes the estimator's stages and data flow; the paragraphs below define each stage.

The estimator is a four-stage composition in the two-step tradition of filtered historical simulation and GARCH-EVT (Barone-Adesi et al., 1999; Hull and White, 1998; McNeil and Frey, 2000). The parametric filter estimates conditional scale, which it does well and with few parameters (Andersen and Bollerslev, 1998; Hansen and Lunde, 2005), and the nonparametric stage estimates the state-dependent shape of the innovation law, which the fixed-shape GARCH-t benchmark fixes by assumption. The parametric route to a state-dependent shape is the autoregressive conditional density model of Hansen (1994), and Stage 2 is its distribution-free counterpart. The extreme-value tail extrapolates beyond the estimation range using the generalized Pareto approximation of Stage 3 below. The conformal stage corrects the level of the quantile curve using its measured miss rate on a held-out split, with the exchangeability-based guarantee of Proposition 1.

Motivation for the design. Conditional scale is identified by a few parameters and estimated with low variance by a parametric filter, so the shape stage is where flexible methods can add value. Per-asset tail data is scarce, which motivates pooling the standardized residuals across names. The EVT stage extrapolates the tail by the generalized Pareto family (Balkema and de Haan, 1974; Pickands, 1975), and the conformal stage is motivated by the split-conformal guarantee of Proposition 1, exact under exchangeability and approximate for the time-ordered residuals used here, with a coverage gap bounded by the average total-variation distance between the residual vector and its swapped versions (Barber et al., 2023). Whether the four stages together improve on the benchmark out of sample is the question of Sections 4 and 5.

Stage 1 (parametric scale). Fit GARCH-t (Bollerslev, 1986, 1987),

$$r_t = \mu + \sigma_t z_t, \qquad \sigma_t^2 = \omega + \alpha(r_{t-1} - \mu)^2 + \beta\sigma_{t-1}^2, \qquad z_t \sim t_\nu \text{ (standardized)}, \tag{2}$$

by quasi-maximum likelihood on a trailing window, and keep $\widehat{\mu}$, $\widehat{\sigma}_t$, and the residuals $\widehat{z}_t = (r_t - \widehat{\mu})/\widehat{\sigma}_t$.

The t_ν likelihood serves as the estimating function for the scale rather than as a belief about the shape, which Stages 2 and 3 re-estimate from the residuals. This two-step design, a parametric filter followed by a residual quantile, is the one whose asymptotics Francq and Zakoïan (2015) derive, and McNeil and Frey (2000) fit the same filter by Gaussian pseudo-likelihood while explicitly declining to believe the innovation law it assumes.

Stage 2 (nonparametric shape). Replace the fixed t_ν innovation quantile with a state-conditioned quantile model of the residuals, trained by pinball ERM *pooled across names*,

$$\widehat{\phi} \;=\; \arg\min_{\phi} \sum_{\text{names } j} \sum_{\text{days } t} \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau\big(\widehat{z}_{j,t} - q_\phi(\tau \mid s_{j,t})\big). \tag{3}$$

Stage 3 (EVT tail). Beyond the estimation range of (3), extrapolate with the generalized Pareto approximation that extreme-value theory motivates for exceedances over a high threshold, under the standard max-domain-of-attraction conditions (Balkema and de Haan, 1974; Pickands, 1975; McNeil and Frey, 2000). The GPD closed form for ES of McNeil and Frey (2000), used below, requires $\xi < 1$, a condition each fitted tail in this paper satisfies by a wide margin.[1] On losses $X = -\widehat{z}$, fix a threshold u at the empirical $(1 - p_0)$ loss quantile with $p_0 = 0.025$. Fit the generalized Pareto distribution (GPD) $G_{\xi,\beta}(x) = 1 - (1 + \xi x/\beta)^{-1/\xi}$ to exceedances $X - u > 0$ strictly on past data. The shape ξ sets how heavy the tail is and the scale β how wide. For $\tau \le p_0$, set

$$\widehat{q}^{\,\mathrm{EVT}}(\tau) \;=\; -\Big[u + \tfrac{\beta}{\xi}\big((\tau/p_0)^{-\xi} - 1\big)\Big], \tag{4}$$

which is continuous at its own endpoint, $\widehat{q}^{\,\mathrm{EVT}}(p_0) = -u$. The formula requires $\beta > 0$ and $1 + \xi(x-u)/\beta > 0$, and reduces to $-[u + \beta \log(p_0/\tau)]$ in the $\xi \to 0$ exponential limit. The threshold u is the *unconditional* empirical loss threshold, while the Stage-2 body quantile at p_0 is the state-conditional $\widetilde{q}(p_0 \mid s)$. The splice is therefore exactly continuous only when the threshold is anchored at the body's own p_0-quantile, $u = -\widetilde{q}(p_0 \mid s)$. Otherwise a small level shift of size $|-u - \widetilde{q}(p_0 \mid s)|$ remains at the join, absorbed by the min-envelope and rearrangement in (5). The tail is applied selectively, where the raw tail under-covers, since it hurts some volatility-index series.

Stage 4 (conformal finish). On a held-out calibration split of size n, compute the signed errors $e_i = \widehat{z}_i - \widehat{q}(\tau \mid s_i)$. Let c_τ be the $\lceil (n{+}1)\tau \rceil$-th order statistic of $\{e_i\}$. The curve is shifted per level by this empirical correction, $\widetilde{q}(\tau \mid s) = \widehat{q}(\tau \mid s) + c_\tau$ (Vovk et al., 2005; Lei et al., 2018), the one-sided form of conformalized quantile regression (Romano et al., 2019). This distribution-free layer repairs the 97.5% level that each model in the comparison set otherwise fails. The coverage statement it inherits is standard (Vovk et al., 2005; Lei et al., 2018; Romano et al., 2019) and is restated here in the form the shift uses.

Proposition 1 (split-conformal validity under exchangeability (standard))**.** *If the calibration and test residuals $e_1, \dots, e_{n+1}$ are exchangeable and almost surely distinct (a continuous error distribution, or randomized tie-breaking), then for each τ with $\lceil (n{+}1)\tau \rceil \le n$ the shifted quantile satisfies*

$$\tau \;\le\; \Pr\left(z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})\right) \;=\; \frac{\lceil (n{+}1)\tau \rceil}{n+1} \;<\; \tau + \tfrac{1}{n+1}.$$

Proof sketch. Under exchangeability and no ties the rank of e_{n+1} among $\{e_1, \dots, e_{n+1}\}$ is uniform on $\{1, \dots, n+1\}$. The event $z_{n+1} \le \widetilde{q}(\tau \mid s_{n+1})$ equals $\{e_{n+1} \le e_{(\lceil (n+1)\tau \rceil)}\} = \{\text{rank} \le \lceil (n{+}1)\tau \rceil\}$, whose probability is $\lceil (n{+}1)\tau \rceil/(n+1) \in [\tau, \tau + \frac{1}{n+1})$ (Vovk et al., 2005). The full argument is in the Online Appendix, together with the role of the no-ties hypothesis (with ties the upper bound can fail, e.g. identical residuals give coverage 1) and the boundary convention $c_\tau = +\infty$ when $\lceil (n{+}1)\tau \rceil = n + 1$. The guarantee is marginal, not conditional on s_{n+1}. □

The proposition's scope is narrower than its use. In practice the calibration split necessarily precedes the test period, and time-ordered residuals are not exchangeable. The proposition therefore motivates the construction. The evidence that it works here is empirical: in the exception-test repairs of Section 5, the conformal stage is the only reason any entrant passes Kupiec at 97.5%. The guarantee also attaches to the shifted quantile alone, not to the full curve of (5), whose tail branch takes the more conservative of the shifted and EVT values and is then monotone-rearranged. The exact finite-sample coverage identity need not survive that composition, so the coverage claim at the reported levels rests on these exception tests rather than on Proposition 1. Conformal constructions with validity under dependence exist (Chernozhukov et al., 2018; Barber et al., 2023; Gibbs and Candès, 2021), but we prefer the simple split form plus an empirical audit.

One further implementation gap: the first-stage GARCH is fit on the full pre-test history, so the calibration errors are held out from the shape learner and the EVT tail but not from the filter. The accuracy-layer comparisons never read the calibration split and share their filter with the benchmark, so their information sets match. Only the overlay's shift uses the split. Re-estimating with the filter stopped at the calibration boundary leaves the accuracy advantage intact (DM 5.1 and 5.4). The strict estimator's gap to the full-window benchmark (-0.6, -2.5) matches that benchmark's own margin over its short-window twin (2.8, 3.8), so the gap reflects a quarter less estimation data rather than leakage. The static shift widens (per-name pass 34% against 48%), consistent with its disclosed conservatism. Under the same strict split the adaptive conformal update of Gibbs and Candès (2021) walks the shift from -0.51 to -0.20, lifts the per-name pass rate to 57%, and shrinks the joint-score gap to GARCH-t from DM -3.2 to -1.5 (not significant at 5%). The estimator's annual-refit schedule sits between the static and adaptive poles, and the exception tests decide between them.

The full forecast composes the stages:

$$\widehat{Q}_t(\tau) \;=\; \widehat{\mu} + \widehat{\sigma}_t \cdot \mathrm{R}\Big[\,\min\{\widetilde{q}_\phi(\tau \mid s_t),\, \widehat{q}^{\,\mathrm{EVT}}(\tau)\}\,\mathbb{I}\{\tau \le p_0\} \;+\; \widetilde{q}_\phi(\tau \mid s_t)\,\mathbb{I}\{\tau > p_0\}\Big], \tag{5}$$

where R[·] is the monotone rearrangement across levels (Chernozhukov et al., 2010). The conditional scale $\widehat{\sigma}_t$ carries the day's volatility from the GARCH stage, and the bracketed curve carries the shape from the nonparametric and EVT stages. A forecast is therefore the conditional mean plus scale times the modeled residual quantile: $\widehat{\mathrm{VaR}}_{\tau,t} = \widehat{\mu} + \widehat{\sigma}_t \widehat{Q}^z_t(\tau)$ and $\widehat{\mathrm{ES}}_{\tau,t} = \widehat{\mu} + \widehat{\sigma}_t\, \tau^{-1} \int_0^\tau \widehat{Q}^z_t(u)\, du$, with $\widehat{Q}^z_t$ the bracketed standardized-residual curve. The tail forecast is the pointwise more conservative branch (the body branch is the minimum on 38% of tail nodes), and both risk measures come from this one curve. Recomputing ES_α as the numerical integral of (5) rather than the GPD closed form leaves each comparison in place. At 2.5%, FZ0 is 1.849 against 1.851, the integral marginally better, DM 5.3, and the estimator-over-GARCH DM is 4.8 and 5.1 at the 1% and 2.5% levels. The splice level is immaterial ($p_0 \in \{1.5\%, 2.5\%, 5\%\}$: DM 4.6–4.8).

Industry models as special cases. Equation (5) nests the standard toolkit, in the taxonomy of Kuester et al. (2006). Historical simulation sets $\widehat{\mu} \equiv 0$, $\widehat{\sigma}_t \equiv 1$, and q the unconditional empirical quantile of past returns. Filtered historical simulation (FHS) (Barone-Adesi et al., 1999; Hull and White, 1998) takes q as the unconditional empirical quantile of past *residuals*, i.e. Stage 2 with the state dependence deleted. GARCH-t (Bollerslev, 1987) takes $q = t_\nu^{-1}$, a fixed parametric shape. Whatever FHS captures, (3) can represent, and it adds conditional shape. CAViaR (Engle and Manganelli, 2004) sits outside the nesting. It autoregresses on the quantile directly, with no scale/shape split, and is the benchmark with which the estimator ties (Section 5).

Algorithm 1 (residual-hybrid estimator). (i) Fit (2) per asset on a trailing window. Store $\widehat{\sigma}_t$, $\widehat{z}_t$. (ii) Pool $(s_{j,t}, \widehat{z}_{j,t})$ across names. Train the shape model by (3). (iii) Fit the GPD tail (4) on past exceedances. Splice at p_0. (iv) Learn the conformal shifts c_τ on a calibration split disjoint from the training data. (v) Forecast by (5). In *ongoing use* all stages refit on an annual walk-forward schedule. The paper's *evaluated* protocol freezes each stage once per split, the more conservative test since no stage ever adapts to the test era. Section 3.1 verifies the frontier separately under the true annual-refit schedule.

[Figure 1 about here.]

2.3 Estimating the shape: gradient boosting and neural quantile generation

Two estimators realize $q_\phi(\tau \mid s)$ in (3). We use trees for point accuracy and the network when draws are needed (Online Appendix Table OA.3).

Gradient-boosted trees (GBM). For each level τ_k on a fixed grid, an additive ensemble of regression trees $q^{(m)}(\cdot) = q^{(m-1)}(\cdot) + \eta\, h_m(\cdot)$ is grown stagewise by gradient boosting (Friedman, 2001). Quantile regression forests (Meinshausen, 2006) are the other tree-based route and are not used here. Each new tree h_m is fit to the negative gradient of the pinball loss, which equals τ_k where the current model under-predicts and $\tau_k - 1$ where it over-predicts. The fitted grid is made non-crossing by monotone rearrangement, which sorts the values $\{q(\tau_1|s), \ldots, q(\tau_K|s)\}$ in τ and is weakly closer to the true monotone quantile curve in every L^p norm (Chernozhukov et al., 2010). On financial tabular state vectors this is the strongest point estimator, about 0.75% better pinball than the tuned network, consistent with the broader tabular-learning literature.

The neural implicit quantile network (IQN), the GBC estimator. A τ-conditioned network in the factorized form of Dabney et al. (2018) and Polson and Sokolov (2023),

$$H_\phi(s,\tau) \;=\; g\big(\psi(s) \circ \varphi(\tau)\big), \qquad \varphi_j(\tau) = \mathrm{ReLU}\Big(\textstyle\sum_{i=0}^{n_c-1} \cos(\pi i \tau)\, w_{ij} + b_j\Big), \tag{6}$$

where $\circ$ is elementwise multiplication and g, ψ are feed-forward nets. Three modifications make it competitive on financial data. The first is tail-aware τ-sampling, $\lambda = \frac{1}{2}\,U(0,1) + \frac{1}{2}\,\mathrm{Beta}(0.3, 0.3)$, which oversamples the extremes that uniform sampling starves (the raw net's 99% breach rate was 3.2%). The second is monotone rearrangement of the τ-curve, and the third is the conformal layer of Section 2.2. The network's role is generation rather than point accuracy: with $\tau \sim U(0,1)$, $z^* = H_\phi(s, \tau)$ draws from the fitted conditional law by inverse transform. The trade-off between the two estimators is set out under *Estimator choice* below.

How training works. The network minimizes the empirical version of (3) by stochastic gradient descent, each example drawn at a fresh level $\tau_{jt} \sim \lambda$ each epoch, so one set of weights carries the entire quantile curve. Early stopping is on held-out pinball loss. Monotone rearrangement and the conformal shift are applied *after* training, and the whole pipeline is refit on the annual walk-forward schedule using only past data. Layer sizes, learning rates, and seeds are reported in the replication package.

Amortization. One model is trained over the pooled (state, loss) pairs of hundreds of names. In generative-Bayes terms, H is an amortized map evaluated at any new (s, τ) without per-asset refitting. It works, at cold-start in particular, because the mapping is estimated across names rather than within one. A per-asset GARCH learns *name-specific parameters* from that name's own history, so with no history it has nothing. The amortized model instead learns a single mapping from observable state to residual shape, estimated from millions of asset-days across names. A newly listed asset is therefore not a new problem: its first ticks and characteristics place it at a point of the state space already densely populated by other names' histories, and the estimated mapping applies there directly. As the name's own history accrues, its trailing realized volatility enters s_t and the forecast tracks its own dynamics automatically. For the same reason the explicit own-history blends add nothing: the state vector already carries the name's own recent information. Feature ablation at full scale (1.32M rows, 720 names, 288 held out) shows the transfer edge is carried overwhelmingly by each name's own recent dynamics, realized vol above all. Removing it costs +1.0%, while removing all cross-sectional characteristics costs only +0.5%. Characteristics matter at cold-start. Two explicit hierarchical corrections, a naive quantile blend toward own-history and an amortized-as-prior affine Gibbs update (Jiang and Tanner, 2008; Bissiri et al., 2016), both hurt at each listing age (ratios 1.003–1.043): rich conditioning performs the partial pooling implicitly. Two external checks: on held-out names the transfer win rate over own-history benchmarks is 59–79%, and the same pipeline scores a leakage-safe weighted scaled pinball of 0.269 (benchmark tier) on the M5 competition data, outside finance altogether.

Estimator choice by forecasting objective. The two estimators minimize the same loss over different index sets. The trees solve $\min \sum_k \sum_i \rho_{\tau_k}(\widehat{z}_i - q_k(s_i))$ on a fixed grid $\{\tau_k\}$, one model per level, while the network solves $\min_\phi \sum_i \mathbb{E}_{\tau\sim\lambda}\, \rho_\tau(\widehat{z}_i - H_\phi(s_i, \tau))$, one model for the continuum. The choice follows from what the application consumes. At fixed regulatory levels (VaR/ES reporting, limits, capital) the trees are preferable: best point accuracy, no GPU, no sampling. Where the application consumes draws (scenario generation, portfolio aggregation by simulation), the network is preferable. It learns one continuous conditional quantile map, so draws come by inverse transform with no grid interpolation, at a measured cost of $\sim$0.75% in point accuracy. A GARCH-t or FHS model also simulates, from its fixed innovation law, whereas a tree grid requires monotone interpolation first. If both are needed, run both from the same pooled residual panel. They share each upstream and downstream stage of (5), so swapping one for the other changes a single component of the pipeline.

The amortized estimation loop is stated as Online Appendix Algorithm OA.1.

2.4 Formal results: why shape misspecification must show up in the loss

This paper's organizing claim, that the nonparametric edge is concentrated where the trailing residual-shape diagnostics are extreme, is an empirical finding with a structural motivation in the pinball loss. The results below are modest: a shape violation creates a quantile wedge, and regret is locally quadratic in that wedge. They motivate looking where trailing shape diagnostics are extreme, but they do not prove that mk_{63} (Section 2.6) estimates the wedge. That link is established empirically. The identity in Lemma 1 is the quantile-score divergence of Gneiting (2011), in the mixture form of Ehm et al. (2016), and the lower bound in (8) is the self-calibration inequality of Steinwart and Christmann (2011). We restate both because Proposition 2 uses the constants.

Lemma 1 (pinball regret identity (standard))**.** *Let $Z \sim F$ with $\mathbb{E}|Z| < \infty$ and $q_F = F^{-1}(\tau)$, and let $S(q) = \mathbb{E}_F\, \rho_\tau(Z - q)$ be the expected pinball loss of forecasting the constant q, finite by the moment condition. Then for any q,*

$$S(q) - S(q_F) \;=\; \int_{q_F}^{q} \big(F(u) - \tau\big)\, du \;\geq\; 0, \tag{7}$$

with equality iff $F(u) = \tau$ for Lebesgue-almost each u between q_F and q. If F has density bounded as $0 < m \leq f \leq M$ on that closed interval,

$$\tfrac{m}{2}\,(q - q_F)^2 \;\leq\; S(q) - S(q_F) \;\leq\; \tfrac{M}{2}\,(q - q_F)^2. \tag{8}$$

Proof. In the Online Appendix. □

Lemma 1 says the extra loss a misspecified forecaster pays is *locally quadratic in its quantile error*. The frontier claim then reduces to whether shape misspecification creates a quantile wedge at the risk levels that matter. For the paper's canonical case, a fixed Student-t shape when the true local shape is fatter, the premise of GARCH-EVT (McNeil and Frey, 2000; Francq and Zakoïan, 2015) and the pattern in the horse race of Kuester et al. (2006), it must:

Proposition 2 (tail wedge under tail-shape misspecification (elementary))**.** *Let $Q_\nu(\tau)$ denote the τ-quantile of the unit-variance Student-t with $\nu > 2$ degrees of freedom. For each pair $\nu_2 > \nu_1 > 2$ there exists $\tau^*(\nu_1, \nu_2) \in (0, 1/2]$ such that $Q_{\nu_1}(\tau) < Q_{\nu_2}(\tau)$ for all $\tau < \tau^*$. The fatter-tailed law thus has the strictly more extreme tail quantile, and the wedge $|Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)| \to \infty$ as $\tau \downarrow 0$. Consequently, by Lemma 1, a forecaster committed to shape t_{ν_2} while the local truth is t_{ν_1} pays regret bounded below by $\frac{m}{2}\left(Q_{\nu_1}(\tau) - Q_{\nu_2}(\tau)\right)^2$ at each level $\tau < \tau^*$, where m is the minimum of the true density on the wedge interval.*

Proof. In the Online Appendix. □

Remark 1 (the crossover level). Unit-variance fat-tailed laws put more mass near the center as well as in the far tail, so the quantile ordering of Proposition 2 reverses at moderate levels. Numerically, the unit-variance t_3 has a less extreme 5% quantile than the normal (-1.36 vs -1.64) but a more extreme 1% quantile (-2.62 vs -2.33). The sign flip occurs at a single crossover level $\tau_c \approx 0.0179$ (solving $Q_{t_3}(\tau) = \Phi^{-1}(\tau)$ numerically for the lower-tail root). The deepest regulatory level, $\tau = 0.01$, therefore sits below the crossover, where the fat tail is strictly more extreme. The level $\tau = 0.025$ sits just above it: at 2.5% the unit-variance t_3 quantile is -1.837, still less extreme than the normal's -1.960. Two of the paper's design choices are direct consequences. First, CRPS-style whole-distribution scoring can rank a fat-tail-aware model below a thin-tailed one even when the tail is the decision-relevant region (Section 2.1), because body and far tail disagree in sign, the weighting problem that Gneiting and Ranjan (2011) address and an instance of the score dependence of rankings under misspecification in Patton (2020). Second, the deepest regulatory level straddles the crossover rather than sitting cleanly beyond it, and the score of Section 2.6 is used as a rank rather than a moment estimate.

The ordering in Proposition 2 is textbook regular variation (Embrechts et al., 1997), and its content is the pairing with Lemma 1. The tail-wedge result motivates, but does not on its own establish, the empirical regularity that organizes the paper: that the nonparametric edge is ordered by the misspecification score. We do not claim a monotonicity theorem at the regulatory levels, because two features of the standardized problem block one. First, Proposition 2 orders a given pair of tail indices only below a pair-specific level $\tau^*(\nu_1, \nu_2)$. It does not deliver monotonicity of the wedge in the tail index at a fixed operational τ. Second, the innovations are unit-variance by construction, and for unit-variance laws the level-τ quantile is not monotone in tail heaviness. As the tail fattens, the variance normalization pulls the fixed-τ quantile back toward the center (the crossover of Remark 1), so that at $\tau = 0.025$ a heavier-tailed t can carry the *less* extreme quantile. We therefore treat the score-ordering as an *empirical* regularity, documented below across the design panel, the frozen holdout, the FRTB comparison, and the cross-market universes, and motivated by the tail-wedge mechanism rather than implied by it. The one exact statement we lean on is the regret identity (Lemma 1), the elementary basis for the consistency of the pinball loss (Gneiting, 2011; Ehm et al., 2016; Steinwart and Christmann, 2011). Wherever the benchmark's reported quantile differs from the oracle's at a level, it pays a pinball regret that is positive and, to second order, quadratic in that gap. The score is our observable proxy for where that gap is largest.

Lemma 2 (validity of the generative sampler (standard))**.** *If $\tau \mapsto \widetilde{q}(\tau \mid s)$ is nondecreasing and left-continuous (guaranteed by the rearrangement step) and $\tau \sim U(0,1)$, then $Z^* = \widetilde{q}(\tau \mid s)$ has quantile function exactly $\widetilde{q}(\cdot \mid s)$.*

Proof. In the Online Appendix. □

Remark 2 (nesting and ERM dominance). The hypothesis class of Stage 2 contains the constant-in-s map $q(\tau)$, i.e., FHS (Barone-Adesi et al., 1999). Empirical risk minimization over the larger class therefore attains in-sample pinball risk no worse than FHS by construction, and the out-of-sample comparisons of Section 5 test whether that dominance survives estimation noise. It does. The same argument does not apply to CAViaR (Engle and Manganelli, 2004), so CAViaR is the one rival the estimator only ties.

2.5 Comparison with generative Bayesian computation

Relative to the generative Bayesian computation of Polson and Sokolov (2023), the downside-risk target changes the estimator in five ways. The sampling measure is tail-weighted, $\lambda = \frac{1}{2}U(0,1) + \frac{1}{2}\text{Beta}(0.3, 0.3)$ in place of $\tau \sim U(0,1)$, because uniform sampling starves the extreme levels that downside risk depends on. The object modeled is the GARCH-standardized residual rather than the raw return, so that the conditional-scale dynamics the parametric model already estimates well are kept. Beyond the training support, extrapolation is handed to the GPD tail (4) spliced at p_0, so that it is governed by extreme-value theory and not by the network. The conformal stage adds the exchangeability-based coverage statement of Proposition 1. Finally, the amortization runs across the real cross-section of hundreds of assets instead of across simulations of a single model, which is what produces the cold-start forecasts and transfer results of Section 2.3. The question changes as well. When such machinery improves on the parametric filter is not something generative computation addresses, and the score of the next subsection is built to answer it.

2.6 The misspecification score

We use "misspecification score" as shorthand for a diagnostic of local residual-shape stress under the fitted GARCH-t benchmark. It is a trailing-window moment monitor in the spirit of the PITbased density-forecast diagnostics of Diebold et al. (1998) and Berkowitz (2001), used as a rank rather than as a formal specification test.

GARCH-t assumes standardized residuals $z_t = (r_t - \mu_t)/\sigma_t$ are i.i.d. Student-t with fixed shape (Bollerslev, 1987). Static heavy tails do not violate this, since ν absorbs them. A locally wrong shape does: recent residuals far fatter-tailed or more asymmetric than any fixed t, the time variation in conditional skewness and kurtosis that Hansen (1994) models parametrically. The score is therefore computed per asset-day over the trailing window $W_t = \{t-63, \ldots, t-1\}$ of $n_w = 63$ residuals, with window mean $\bar{z}$ and standard deviation s_z. The window *ends the day before the forecast target*, so each signal is lagged at least one day relative to the loss it is paired with. The frontier sorts of Section 4 pair S_{t-1} with the day-t loss, an alignment enforced by construction. The two moment statistics are

$$\mathrm{mk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^4 - 3, \qquad \mathrm{sk}_{63}(t) = \frac{1}{n_w} \sum_{i \in W_t} \left(\frac{\widehat{z}_i - \bar{z}}{s_z} \right)^3, \tag{9}$$

In estimation, mk_{63} and sk_{63} are computed as the bias-corrected sample excess kurtosis and skewness. At the full window length the bias correction is a common affine rescaling of (9), so the pooled decile ranks that the frontier uses are unchanged. *Skewness* is the signed third moment sk_{63}. *Asymmetry* always means its absolute value $|\mathrm{sk}_{63}|$, departure from symmetry in either direction, since a fixed symmetric t is violated equally by either sign. The score also uses the five-day jump indicator $J_5(t) = \max_{i \in \{t-5,\ldots,t-1\}} |\widehat{z}_i|$. The frontier orders the edge by the decile of each of these signals. The composite score is the maximum of their trailing percentiles, $\mathrm{score}_t = \max\{\widehat{F}_t(\mathrm{mk}_{63}), \widehat{F}_t(|\mathrm{sk}_{63}|), \widehat{F}_t(J_5)\}$, where $\widehat{F}_t$ is each signal's percentile in the trailing panel available before t. The composite is itself deciled. All three are predetermined (trailing-window) and computable on any asset with a fitted GARCH, so the frontier can be monitored in real time.

Online Appendix Table OA.3 reports the associated per-decile actions.

Why these statistics. Conditional on the fitted location-scale specification (μ_t, σ_t), the distributional restriction the flexible family relaxes is the fixed t_ν innovation *shape*. The lowest-order ways a fixed symmetric law can be locally wrong are the third and fourth standardized moments (Bollerslev, 1987; Hansen, 1994). The statistic sk_{63} detects asymmetry no symmetric t can carry, and mk_{63} detects tail mass beyond what the fitted ν implies. An exact t_ν sample has known positive excess kurtosis, but the score is used as a rank, so only its cross-sectional and temporal variation matters, not its t_ν baseline. J_5 catches jump episodes too recent for the window moments to register. When these are elevated, the ERM shape model (3) has something to learn that t_ν^{-1} cannot represent, and Section 4 shows that the realized edge concentrates there.

The monitoring procedure is stated as Online Appendix Algorithm OA.2.

3 Data and evaluation protocol

Equity panels come from CRSP (the Center for Research in Security Prices) daily files accessed through WRDS. The design panel covers 2014–2024 and contains the 200 names with the deepest continuous histories among those with at least 1,500 daily observations. Each modeling choice in the paper was made on this panel. The untouched-era holdout covers 2000–2013 and was pulled once, after the specification was frozen: the top 200 common shares (share codes 10 and 11) by average market capitalization among those with at least 3,000 observations in the window. Returns are in percent and include CRSP delisting returns where present. A name whose delisting return CRSP does not record ends at its last regular return, the omission Shumway (1997) shows is large and negative for performance-related delistings. The one delisting event in the design panel moves the main results by less than two basis points. The cross-asset sweep (43 instruments), the country-index panel (26 markets), and the single-asset FRTB fits use Bloomberg terminal data.

The replication repository carries code, configuration, and derived statistics, and the panels can be rebuilt from the queries above by any licensed subscriber. Table 1 gives the sample flow.

[Table 1 about here.]

Splits are chronological within each name. The first 60% of a name's history is the estimation window and the last 40% is test. The final quarter of the estimation window (15% of the history) is held out as the calibration split. A GARCH-t model is fitted once per name on the estimation window and its variance recursion is then filtered through the full history, so test-period scales depend on test-period data only through the recursion itself. The pooled residual quantile model is fitted on pooled estimation-window residuals, the GPD tail on the same pooled residuals below their empirical 2.5% point, and the conformal shifts on the pooled calibration split. No stage refits inside the test window unless a rolling variant is named explicitly. Two structural properties of this protocol, calendar overlap in pooled training and the universe rule's survivorship, are examined in their own subsections at the end of this section.

Loss and test conventions are collected in the note that closes this section. Distributional accuracy is scored by mean pinball loss across twelve quantile levels, a quantile-weighted proper score (Gneiting and Ranjan, 2011), and joint (VaR, ES) pairs by the zero-homogeneous FZ loss (Fissler and Ziegel, 2016; Patton et al., 2019). Loss comparisons use per-date Diebold–Mariano statistics (Diebold and Mariano, 1995) with a Newey–West variance (Newey and West, 1987), and multi-model fields use the Model Confidence Set (Hansen et al., 2011). Exception behavior is tested by Kupiec, Christoffersen, and date-clustered breach tests at the two regulatory levels.

"Frozen-specification" has a specific meaning here: a run is frozen-specification when the full specification (signals, features, hyperparameters, split rule, and test statistics) is fixed and the predictions are written down before the data are pulled. We do not use the word *registered*, since no third-party registry or timestamp exists for them. The holdout pipeline is line-for-line the design-era pipeline, and both are public in the replication repository, so the two can be compared directly to confirm that no signal, feature, hyperparameter, or test statistic was altered for the holdout era. The 2000–2013 holdout of Section 4 ran under this rule. As a post-hoc check, the composite score also replicates on this holdout. Its top decile carries +1.12% (DM 2.51), marginally above its kurtosis component (+1.04%, DM 2.16), and +0.71% (DM 2.08) under the expanding prior-only cutoff. Because the composite was not the rule frozen at the time, this is replication on an already-examined sample, not a frozen-specification test.

3.1 Common shocks, calendar overlap, and the pooled learner

Two nested flexible specifications appear in the paper. The residual-hybrid holds the GARCH conditional scale fixed and models the standardized residuals; it produces the design-panel frontier (Section 4) and every VaR and ES forecast. A simpler returns-space variant, plain enough to fix in advance, carries the pre-committed out-of-sample and robustness checks. Because a 60/40 per-name split can place one name's training window on another's test dates (López de Prado, 2018), a pooled learner could meet a systemic event in training that a per-name model meets only out of sample. A strict calendar-time split closes that channel (Gu et al., 2020): each stage is re-estimated strictly before 2020-01-01 and tested after, and the top-decile edge survives (+2.47%, DM 6.22). It also survives a pre-2007 cut that removes the 2008 crisis from every training window (+0.29% overall, DM 3.08), and an annual walk-forward that refits the GARCH, residuals, score, and learner at each January-1 cutoff, under which the top decile still carries +2.09% at DM 4.43 while the average edge washes out (−0.18%). Stale benchmark parameters therefore do not produce the conditional result. The overlap mechanism and the full calendar-split, pre-2007, and walk-forward tables are in the Online Appendix.

3.2 Cross-sectional dependence, family-wise error, and the universe rule

Each main test collapses the cross-section to one number per date before any variance is estimated, and the bootstraps resample dates rather than asset-days, following the pooled-average panel test of Qu et al. (2024). Over the full 3×10 signal-decile grid, Romano–Wolf step-down family-wise control (Romano and Wolf, 2005, 2016) leaves nine cells significant at 5%, all of them frontier cells and including the top deciles of all three signals; any cell that does not survive is reported as indistinguishable from zero wherever it is discussed. The exploratory single-instrument universes carry their own Holm adjustment over 93 comparisons (Holm, 1979). Survivorship is addressed by a point-in-time re-selection that fixes the universe at the top 200 names as of Q1-2000, applies no minimum-history filter, and rides each delisting return to the end: on the 164 names that clear the fitting floors the overall edge is +0.58% (DM 8.96) and the frozen top bucket carries +2.94% at DM 6.6, larger than in the survivor-tilted holdout. The date-bootstrap construction, the cell-by-cell adjustment, and the point-in-time details are in the Online Appendix.

3.3 Reporting and inference conventions

The reporting and inference conventions used in the empirical sections that follow are fixed here. Edges are percentage reductions in pinball loss, so larger is better for the named model. A "+2.7% edge" means the model's average loss is 2.7% lower than the benchmark's. Ratios are model-over-benchmark loss ratios, so below 1 favors the model and $1 -$ ratio is the edge ($0.973 =$ a 2.7% win, $1.043 =$ a 4.3% loss). DM is the Diebold–Mariano t-statistic on the loss difference. The forecasts are frozen objects, so the comparison is of forecasts rather than of estimated models, the use Diebold (2015) endorses. For the per-date DM, losses are first averaged across names within each date, $d_t = N_t^{-1} \sum_i (\ell_{it}^A - \ell_{it}^B)$. The orientation is stated at each use (benchmark minus the residual-hybrid in its main comparisons, so that positive favors the residual-hybrid). The test is then run on the resulting ∼1,100-date series with a Newey–West (lag-10) variance (Newey and West, 1987), so the effective sample is the number of dates, not the number of asset-days. The lag replaces the $h - 1 = 0$ truncation of Diebold and Mariano (1995) because misspecified one-step forecasts leave serially correlated loss differentials. Reported p-values are one-sided in the direction of the named model ($t > 1.645$ at 5%, $t > 2.33$ at 1%), a direction fixed before the run for the residual-hybrid's main comparisons, and where a winner is instead chosen by the data the one-sided level reads as a two-sided level of twice its size. A t-statistic above 2.6 is significant under either convention, and marginal cases are flagged where they occur. A tie means the per-date DM cannot reject zero at the two-sided 10% level. The point estimate is reported with the statistic so the reader can judge the width, and failure to reject is not treated as proof of equivalence. Where a genuine equivalence statement is possible we make it as one. For the CAViaR comparison the 90% confidence interval for the loss difference, $[-0.05\%, +0.11\%]$ of the benchmark loss, lies wholly inside a $\pm 0.25\%$ margin, which is a two-one-sided-tests conclusion rather than a failure to reject. The Model Confidence Set is computed on the same per-date loss matrix with the range statistic of Hansen et al. (2011) and a stationary bootstrap over dates (Politis and Romano, 1994) (mean block length 10 days, $B = 1{,}000$ replications in the main comparison, $B = 800$ in the CAViaR extension). Hansen et al. (2011) implement the set with a fixed-length block bootstrap (Künsch, 1989), and the stationary variant is the one used in the SPA test of Hansen (2005). Coverage numbers are realized frequencies against a stated nominal (0.05 target: 0.056 is slight under-coverage of risk, 0.032 over-conservatism). For credible-interval coverage, closer to nominal is better in both directions. Correlations quoted for the frontier relate a universe's residual-kurtosis level to the size of the nonparametric edge across that universe's members. No other significance marks are used.

4 The misspecification frontier

4.1 The within-universe result

On 200 CRSP names across 1,108 test dates (≈221.6k asset-days), we bucket the per-day pinball edge of the residual-hybrid over GARCH-t into deciles of each misspecification signal. Both forecasters share the GARCH conditional scale, so any advantage enters through the residual quantile. Because that quantile is conditioned on trailing realized volatility as well as on the score, it can also act as a state-dependent correction to the daily scale, and Section 5 shows that on large capitalizations a realized-variance scale absorbs much of this component (Andersen and Bollerslev, 1998; Hansen and Lunde, 2005). The frontier therefore measures the value of a state-conditioned residual law given a daily parametric scale, not a pure innovation-shape effect. Sorting forecast performance by a predetermined state is a conditional predictive-ability comparison in the sense of Giacomini and White (2006). Table 2 reports the sort on which the rest of the paper builds.

[Table 2 about here.]

The edge is concentrated in the top decile: the top mk_{63} decile carries +2.98% at per-date DM 10.5, the overall edge is +0.35% at DM 5.35, and the lower deciles are statistically indistinguishable from zero.

The composite score. Table 2 sorts on each signal separately. The monitor the paper uses is the composite score, the maximum across the three signals' trailing-panel percentiles (kurtosis, asymmetry, jump). Evaluated directly, its top decile is a 10% cell carrying +2.79% (DM 8.9), and the lower nine deciles sit between −0.3% and +0.35%. The edge survives the strictly non-anticipating expanding cutoff at +2.72% (DM 9.0). The three components each carry a top-decile edge on their own: +2.98%, +2.78%, and +1.96% (kurtosis, asymmetry, jumps). The maximum of decile ranks instead flags the union of the three top deciles, a 19% cell at +1.72%, so we report the re-deciled composite.

The score's content could reflect model-relative misspecification, departure from each name's own fitted t_{ν_i}, or the absolute level of recent tail activity. Converting each mk_{63} into a percentile of the simulated null of 63-window kurtosis under the name's fitted t_{ν_i} leaves the frontier essentially unchanged (top decile +3.0%, DM 8.5, against +3.0%, DM 10.5 raw). The normalization is nearly rank-preserving, however (Spearman 0.97, 82% top-decile overlap), so it cannot by itself separate the two. A discriminating test instead favors the raw-tail-activity reading. In a within-date cross-sectional (Fama–MacBeth) regression of the per-observation edge on the raw score and the ν-relative percentile, the raw score carries the predictive content ($t = 7.8$), while the ν-relative percentile has significantly negative incremental content ($t = -5.2$). Double-sorting agrees. The top-raw-kurtosis-quintile edge falls from +1.9% for the most fat-tailed (low-ν) names to +1.5% for the thinnest-tailed, and observations high in the ν-relative score but not in raw kurtosis carry no edge (+0.3%, DM 0.5). The signal is thus the absolute level of recent tail activity rather than a per-name ν-normalized surprise. The signal still measures shape stress relative to the fitted model, since mk_{63} is computed on the standardized residuals the model treats as stationary t_ν. Elevated values flag states in which the realized residual shape is heavier or more asymmetric than the fitted law typically produces. A correctly specified low-ν t_ν can itself generate high realized kurtosis, which is why the score is treated as an indicator of stress and makes no claim to be a specification test (Section 2.6). The data reject the finer per-name normalization. A plausible reason is that a name's fitted ν is estimated from the same residuals and a low value is itself a symptom of the tail activity that matters, so dividing it out removes part of the signal.

Scale versus shape. The top decile follows large standardized residuals, and since a GARCH(1,1) inflates its conditional variance after such a shock, part of the edge reflects the standard filter's post-outlier scale error rather than any error in the innovation shape. The two can be separated with a jump-robust GARCH, which bounds the squared-innovation news term at $K\sigma_{t-1}^2$ (a $\sqrt{K}$-sigma cap that keeps a single shock from propagating into the variance) and so does not overshoot after a single shock. Against that benchmark the top-decile edge falls from +2.98% (DM 10.5) to +0.45% (DM 4.7) at a three-sigma cap and to +0.27% (DM 2.5) at four sigma, while the robust filter alone accounts for +3.17% (DM 8.2) of the standard-GARCH gap in that decile. Most of the frontier is therefore daily-scale misspecification, which a robust scale corrects on its own; what remains under a robust scale is a conditional-shape edge of the order of a third to a half of a percent, smaller but still significant. The large-cap realized-variance decomposition of Section 5 tells the same story. It is the reason the score is read as a monitor of parametric-model stress, part scale and part shape, and not as a diagnostic of shape alone.

Robustness of the frontier. Four checks, in the Online Appendix, rule out the obvious artifacts. The one-day score lag is not a look-ahead window: it gives the largest top-decile edge, and a five-day-old score keeps most of it (Online Appendix Figure OA.1). Re-forming the decile cutoff from strictly earlier dates, the rule the monitor uses, leaves 86% of top-decile membership unchanged and the edge intact (+2.45%, DM 8.3), so the concentration is not an artifact of the pooled-panel sort. A per-name FHS reweighted toward days of similar mk_{63} is significantly worse than plain FHS (per-date DM -11.3), so a parametric-adjacent fix cannot absorb the edge and the pooled learner is doing the work. A double sort on mk_{63} and trailing volatility shows the edge needs both a shape break and an active regime, peaking at +2.33% where both are present (Online Appendix Table OA.2). A superior-predictive-ability test (Hansen, 2005) against the industry benchmarks returns $T^{\text{SPA}} = 0$, and with the design-era threshold held fixed the top-decile edge is positive in every test year from 2020 to 2024.

The untouched-era holdout. Each design choice in this paper was made on data from 2014 onward. As a guard against specification search, we froze the entire pipeline (signals, features, hyperparameters, split rule, and test statistics) and wrote down three predictions in advance: a positive and significant top-decile edge with a bottom decile near zero, a small positive overall edge, and a threshold-shaped decile profile matching Table 2. We then pulled CRSP daily returns for 2000–2013, an era not used at any earlier stage of the analysis. The pre-specified kurtosis-component frontier, a simpler returns-space variant frozen in advance, replicates: the top-decile edge is +0.89% with per-date DM 2.16 ($p = 0.016$, 1,465 dates), deciles one through nine sit between −0.18% and +0.28%, and the overall edge is +0.02%. The residual-hybrid composite replicates as well (Figure 2): its top decile carries +1.12% at per-date DM 2.51. This is the same threshold nonlinearity, attenuated in a larger-cap, pre-2014 universe. The decile boundaries above are within-sample ranks of the holdout test era, the standard conditional-sort convention. The partition rule never touches a forecast or a loss, but it could not have been applied in real time on day one. A day-one-applicable restatement freezes the design era's numeric decile boundaries (the pooled design test sample's mk_{63} edges, top threshold 13.67) and applies them to the holdout as constants. Membership at (i, t) then depends only on name i's own past 63 days and a number fixed before any holdout day was scored. The threshold shape survives the frozen rule: bins eight through ten carry +0.82%, +0.92%, and +1.02% against −0.3–0.0% below, and the top-bin point edge slightly exceeds the within-sample sort's +0.89%. The cell thins to 3.3% occupancy in this lower-kurtosis era, however, and the per-date test loses power (DM 1.11, insensitive to Newey–West lags 5–44). A second real-time rule needs no design-era constant at all. An expanding pooled 90th-percentile threshold over strictly earlier dates (250-date burn-in) restores natural occupancy (9.5%) and recovers the sort's result with significance (+0.99%, DM 2.15, insensitive to Newey–West lags 5–44). The bottom bucket is -0.18% ($t = -1.4$). The conditional sort, the frozen constant, and the recursive rule agree on the size of the top-cell edge, and the recursive rule is the version that could have been run in real time from day one. The frozen thresholds are also tested on the point-in-time universe of Section 3.2, where they carry +2.94% at DM 6.6.

[Figure 2 about here.]

The overall signal–edge correlation is low (0.05–0.10) because the relationship is a threshold non-linearity, so the top-decile Diebold–Mariano statistics are the appropriate summary.

4.2 The frontier across universes

[Table 3 about here.]

Table 3 shows the same axis at work across markets. The decisive standalone wins are the hyperinflation and capital-control currencies, the Argentine peso and the Egyptian pound, 12–14% better and individually significant, whose residual kurtosis (∼2000–3300) sits two orders of magnitude above anything GARCH-t was built to absorb. Twenty-six country indices trace a gradient that tracks residual kurtosis rather than the development label, and forty-three cross-asset instruments extend the pattern into rates, commodities, and credit. The Korea-2026 crash is the negative control: a price crash under circuit breakers censors the return but leaves residual kurtosis ordinary, and GARCH-t correctly wins all twenty-three assets, so the frontier tracks residual shape and not price turbulence. The instrument-level entries are exploratory, and under a Holm step-down over the 93 comparisons (Online Appendix Table OA.4) seven survive. The aggregate crisis-tier FX statistic (DM 4.12) and the cross-universe kurtosis–edge gradients are the evidence this section rests on. A frequency dimension is left to future work: the five daily crypto series show no edge, while at hourly frequency the picture inverts.

4.3 Regime dependence and model selection

In the amortized panel the absolute edge grows with turbulence. The GARCH-minus-GBM pinball gap grows monotonically from 0.0059 in the calmest realized-vol quintile to 0.0247 in the most turbulent, roughly fourfold. Because pinball loss is scale-equivariant, part of that growth reflects the higher loss level in turbulent states rather than a larger relative edge, and both nonparametric estimators beat GARCH in each quintile (overall: GARCH 0.6543, GBM 0.6447, IQN 0.6478).

Gating, running GARCH on calm days and switching to the nonparametric model when the score is high, does not improve on continuous use of the flexible model. A classifier gate trained on the residual-shape signals routes 90.5% of days to the nonparametric model and matches always-nonparametric exactly (pinball 0.3987, against 0.4000 for GARCH). Its routing also runs backwards across score deciles: nonparametric wins on high-misspecification days are rare but large, while a classifier targets the frequency of wins rather than the conditional mean of the loss differential. A regression gate on the edge magnitude, the forecast-selection rule of Giacomini and White (2006) (regress the loss differential on the state and switch on the sign of the fitted value), fixes the direction (routing flat ≈ 1.0, predicted-edge correlation $+0.16$) and still converges to always-nonparametric. That outcome is the corollary of Table 2: the conditional mean of the differential is positive where the score is high and indistinguishable from zero elsewhere, so no state offers a detectable payoff to selecting, or to placing combination weight on, the parametric quantile. The per-day oracle gap (4.4%) is unreachable because per-day model advantage is dominated by which day the tail event lands, and that is unpredictable from prior-day state. In practice the amortized setting uses the nonparametric model throughout, since it does not underperform on calm days. The residual-hybrid is the choice where a guaranteed $\geq$GARCH floor is wanted, and the frontier score serves as a monitor (Section 6). Detection latency is small (median zero) and the edge is back-loaded within episodes, so the lag costs little even in the per-name setting.

5 VaR and Expected Shortfall forecast evaluation

5.1 Head-to-head accuracy and significance

Table 4 reports the full comparison on 140 CRSP names (155k observations, 1,108 dates).

[Table 4 about here.]

The stress-era replication. The comparison set was re-run on the pre-committed 2000–2013 holdout panel. Splits follow Section 3, so each name's test window runs from roughly mid-2008 through 2013, the crisis and its aftermath. The EVT-tailed residual-hybrid passes Kupiec at 99% for 94% of names and Christoffersen for 95%, with date-clustered t-statistics of 0.17 at 99% and 0.35 at 97.5%, consistent with nominal at both regulatory levels through the crisis window. Historical simulation, by far the most common method in bank disclosures (Pérignon and Smith, 2010), and EWMA do not survive the same test: historical simulation breaches at 1.53% against a 1% target (pass rate 61%, date-clustered $t = 2.08$) and EWMA at 1.57% ($t = 2.83$). GARCH-t is also well calibrated in this era (94% pass rate), as the frontier predicts. Calibration parity is the norm, and the residual-hybrid's added value in any era is the accuracy edge where the misspecification score is high. The conformal layer, calibrated on pre-crisis data, tilts conservative out of era (breach 2.21% at 97.5%, $t = -1.09$, not significant). The safety margin widens when the calibration era is calmer than the test era, and the error runs in the direction regulators prefer.

Adding SAV-CAViaR, a model absent from banks' standard toolkit but a leading semiparametric academic benchmark, produces a statistical tie: CAViaR 0.3461 vs residual-hybrid 0.3460

(DM 0.4, $p = 0.34$), and the 90% Model Confidence Set contains exactly these two. The CAViaR study's common sample differs slightly from Table 4's, which is why the residual-hybrid's loss level is 0.3460 there against 0.3551 in the table. Rankings, not levels, are comparable across the two runs. The residual-hybrid thus ties CAViaR and significantly beats every model in banks' standard toolkit. The two also differ in what they deliver. SAV-CAViaR models one quantile level per fit, produces no ES (the FRTB metric) without an ad-hoc stage, and needs per-name history, so it cannot be applied to new listings; the residual-hybrid produces the full curve and extends to new listings. Pooled exception rejections are not counted against CAViaR, because pooled asset-day tests ignore cross-sectional dependence and overstate rejection for every model, the residual-hybrid included; the date-clustered restatement is the appropriate reading. Two textbook FHS variants, per-name train-constant and per-name rolling 500-day trailing, were run on the identical panel. Both lose to the residual-hybrid, by DM 5.2 and 5.7 (pinball 0.3562 and 0.3569 against 0.3551), and they bracket the pooled variant used in Table 4. A neural IQN used as the residual model loses to trees in its untuned form (0.3650 vs 0.3551, DM 13.3) and has a thin tail (breach$_{99}$ 3.2%). The tail-aware, monotone, conformally recalibrated version narrows the gap (0.3638 vs 0.3632, DM 3.03) and repairs calibration (breach$_{99}$ 0.92%). Trees remain the strongest tabular estimator, and the tuned IQN is competitive. The ES columns of Table 4 are diagnostic only. Under the exact tail integral, each fat-tailed entrant's predicted ES exceeds the realized mean below its own VaR, by 9% (raw residual-hybrid, rolling FHS) to 22% (pooled FHS). The comparator conditions on each model's own breach set. The joint FZ0 score below is the ES ranking used in this paper.

The joint $(\mathrm{VaR}, \mathrm{ES})$ **score.** Because ES is not elicitable alone but the pair $(\mathrm{VaR}, \mathrm{ES})$ is jointly elicitable under the Fissler–Ziegel class (Fissler and Ziegel, 2016), the comparison set was also scored under the FZ0 joint score

$$L_{\mathrm{FZ0},\alpha}(r,q,e) = -\frac{1}{\alpha e}\,\mathbb{I}\{r \le q\}(q-r) + \frac{q}{e} + \log(-e) - 1, \qquad e \le q < 0, \tag{10}$$

the 0-homogeneous member of the class (Patton et al., 2019). Lower values are better. On the large-cap sandbox subset used to fix the score's orientation, a Gaussian-innovation GARCH is significantly worst at both the 2.5% and 1% levels (DM 5.4 and 6.0): under a jointly consistent score, Gaussian innovations are clearly inadequate. Among the fat-tailed entrants the FZ0 differences on large caps are small ($|\mathrm{DM}| < 1.4$), the frontier's own prediction that the shape edge is flat on well-behaved names. On a high-kurtosis subset the ordering favors the shape-aware forecaster, directionally but not significantly at that width. The pre-specified full-panel run, reported next, settles the magnitudes on the residual-hybrid's own forecasts.

[Figure 3 about here.]

On the full 200-name panel (221.6k test asset-days), VaR and ES are read from the coherent min-envelope curve Q^* and per-date DM inference is used throughout. The residual-hybrid's *accuracy layer*, the EVT-tailed variant with no conformal shift, attains the lowest FZ0 at both regulatory levels. At 1% it beats GARCH-t (DM 4.8) and FHS (DM 4.9), the two standard benchmarks the accuracy claim rests on. At 2.5% it again wins over GARCH-t (DM 5.1). The two FZ-estimated dynamic $(\mathrm{VaR}, \mathrm{ES})$ models a referee would turn to next are the one-factor score-driven GAS of Patton et al. (2019) and the ES-CAViaR of Taylor (2019). Each is estimated per name on the same panel by minimizing the same FZ0 loss, and under a three-start optimizer both converge for all 200 names, so the single-start instability of an earlier attempt does not arise. The accuracy layer beats the GAS at both levels (DM 9.5 at 1%, 8.6 at 2.5%), and it beats the Taylor model at 1% (DM 3.6) while tying it at 2.5% (DM 0.1), the near-tie the fat-tailed benchmarks all show at 2.5%

where the shape edge is thin. This ranking survives the annual walk-forward refit of Section 3.1: re-estimating every stage at each January-1 cutoff and re-scoring the 2020–2024 panel, the accuracy layer keeps the lowest FZ0 at both levels, beating GARCH-t and FHS by DM 3.6 and 2.7 at 1% and 3.1 and 2.6 at 2.5% (one-sided $p < 0.005$). Refitting narrows these margins but does not close them, unlike the average pinball edge, which closes under the same refit. The accuracy layer is thus one configuration that attains the lowest FZ0 at both levels and passes the aggregate date-clustered exception test at 99% and marginally at 97.5%. Per-name diagnostics show residual cross-sectional calibration heterogeneity (84% and 86% pass rates at 99%, Online Appendix Figure OA.2). The one failed test is the pooled 155k-observation Kupiec at 97.5%, which at this sample size rejects very small deviations. The frontier concentrates the joint-score edge as it does the pinball edge: the top misspecification decile carries DM 2.4 at the 1% level against DM 0.2 in the panel bulk. Adding the conformal shift at 97.5%, whose calibration split includes the 2020 crash, widens the band out of era (breach 1.6% against the 2.5% target) and gives back the FZ0 advantage there. The static-overlay variant loses to GARCH-t on the 2.5% joint score (DM -2.0), and the concession is largest in the top misspecification decile, where the static shift's joint-score deficit reaches DM -7.1. The coverage property costs sharpness under a strictly consistent score, and its error direction is conservative (wider bands, fewer breaches than nominal). Where the conservative coverage margin is valued, the shift is kept; where FZ0 efficiency is the criterion, the EVT tail is run alone, and it passes the exception tests on its own (Online Appendix Figure OA.2). Most of the concession reflects a frozen margin. The adaptive form of the shift (Gibbs and Candès, 2021), updated daily from the panel breach frequency, shrinks the margin out of era (from -0.35 to -0.10, breach 1.8% $\to$ 2.1%) and erases the overall concession. The adaptive variant ties GARCH-t on 2.5% FZ0 (DM -0.3). The top-decile cost survives either shift (-4.8 vs -6.0), the price of a margin where the bands are widest. The static shift also worsens per-name coverage (Kupiec$_{97.5}$ pass rate 47%, against 72% unshifted and 69% adaptive), because over-coverage fails the test from the other side. Each "lowest FZ0" statement refers to the accuracy layer.

Direct ES calibration backtests. Because exception counts test VaR and not ES, two direct ES calibration backtests, the McNeil–Frey exceedance-residual test (McNeil and Frey, 2000) and the Acerbi–Székely Z_2 statistic (Acerbi and Székely, 2014), complement the jointly consistent FZ0 loss above. Nolde and Ziegel (2017) and Bayer and Dimitriadis (2022) draw the distinction between backtesting the calibration of an ES forecast and comparing forecasts through a joint score. Both are computed on the residual-hybrid's own min-envelope forecasts over the same 200-name panel, with a stationary block bootstrap over the 1,108 test dates (Online Appendix). The Gaussian filter is worst on each ES metric, and among the fat-tailed entrants no single model dominates each diagnostic. The residual-hybrid has the Z_2 closest to zero at 1% and ties GARCH-t at 2.5%, and the residual-hybrid and GARCH-t are the two models that pass the 1% Kupiec test. On the McNeil–Frey residual, FHS, which targets the empirical tail directly, is closest to zero. The calibration tests show that the residual-hybrid's tail is well calibrated. The comparative result is the FZ0 loss, where the edge over GARCH-t and FHS is significant at both levels.

5.2 The ten-day horizon: a direct multi-day extension

The multi-day results in this subsection come from a direct amortized model, a pooled quantile learner trained on h-day cumulative returns with GARCH-derived features; the residual-hybrid estimator of Algorithm 1 is a one-day architecture, and no ten-day claim in this paper attaches to it. Training labels whose h-day window crosses the split are excluded, so no label contains test-era returns. At the FRTB ten-day liquidity horizon (CRSP 2014–2024) the direct model's pinball edge over $\sqrt{h}$-scaled GARCH grows from $\sim$0.4% at $h = 1$ to $\sim$1.4% at $h = 10$, holds in the 2020–2022 stress window (direct model 1.248 $<$ FHS 1.2605 $<$ GARCH 1.262), and, on the design panel's

h-curve (Online Appendix Figure OA.3), rises from 0.45% at h=1 to 2.1% at h=20 under the same purge.

Tail severity is more consequential. Ten-day ES is the same exact tail integral as in Section 5, with the same model-dependent own-tail caveat. Each ten-day tail over-states its own-tail comparator, GARCH $\sqrt{h}$ most (predicted -21.1 vs realized -15.4) and the direct model least (-20.1 vs -17.3), and the direct model has the better ten-day breach$_{97.5}$ (2.79% vs GARCH 3.30% against the 2.5% target). The temporal split, however, places the whole test window in 2020–2024, and the pre-committed rerun on 2000–2013 shows that the ten-day result is era-dependent. Out of era the two methods tie on accuracy (direct edge +0.49%, date-level DM 0.07, NW lag 10), and the calibration verdict reverses. The $\sqrt{h}$-scaled GARCH-t ES is almost exactly calibrated through the crisis window (predicted -16.6 vs realized -16.6, numerical integral, purged), while the direct model's tail under-states it (-15.9 vs -17.8). The pooled ten-day tail, trained on the pre-crisis segment, did not extrapolate to 2008-scale ten-day losses. The ten-day tail-severity advantage is therefore a 2014–2024 statement, and in a 2008-type era the direct ten-day model should be run with the $\sqrt{h}$-scaled parametric number alongside it as a conservative model-risk envelope. The frontier itself is confirmed on 2000–2013 (Section 4). The pre-specified horizon comparison thus ends with a mixed verdict.

5.3 Cross-asset evidence

Across the 43-instrument sweep the relationship is heterogeneous. The volatility indices (VIX, V2X, VVIX) and Baltic freight post the largest accuracy gains and survive the multiplicity adjustment (Online Appendix Table OA.4), but their calibration is open. VIX's breach-independence is marginal (one-sided $p = 0.041$), and credit and Baltic freight fail it, so all are reported as accuracy edges rather than calibrated wins. An apparent electricity win reversed once the return transformation was corrected for near-zero prices. The GPD tail also hurts some volatility indices, motivating the selective-application rule of Section 2.2. Instrument-level results and diagnostics are collected in the Online Appendix.

The scale–shape decomposition. Against a Realized GARCH (Hansen et al., 2012) fitted on intraday data, the daily-data core concedes FZ0 at both levels (DM 6.2 and 4.5). Re-fitting the shape learner, EVT tail, and score on the realized residuals, however, leaves the score within noise. On large caps the frontier is therefore largely daily-scale misspecification. Where intraday data are unavailable or too sparse to build a realized measure (new listings, illiquid names, many cross-asset instruments), the daily score is the version of the diagnostic that can be computed (Online Appendix).

6 Extensions and implications

The full estimation and forecasting pipeline is stated in the Online Appendix.

Using the score. The score is available *ex ante*: a risk manager computes today's value from today's residuals and uses it for tomorrow's forecast. Its role is to stratify, not to switch models day by day, because the per-day oracle gap is not forecastable from prior-day state and a learned gate collapses to always-nonparametric (Section 4.3). The one setting in which it drives a switch is the per-name standalone model, where nonparametric quantiles underperform on calm days. As a monitor it is cheap, since its input is the trailing residual window the scale stage already produces, and it is a slow state that need not be watched daily. On capital, the residual-hybrid is a one-day single-name VaR/ES forecaster of the kind that feeds an internal model, not the ten-day portfolio-level stressed ES that sets the FRTB charge; the paper makes no capital-release claim, because the predicted-versus-realized ES wedge conditions on each model's own breach set and no defensible arithmetic converts it to capital (Basel Committee on Banking Supervision, 2019). The narrower implication is that the ES-based component of the charge scales with the ES forecast.

Cold-start risk for new listings. The amortized model in its characteristics-only form is the only entrant here that can produce a conditional risk forecast for an asset with no return history. An IPO or newly listed ETF gets day-one quantiles from its characteristics and first ticks, because the model was trained on the pooled (state, loss) pairs of hundreds of other names; the residual-hybrid of Section 2.2 attaches once a scale filter is estimable. The amortized forecast beats own-history benchmarks at each listing age, and the edge is largest when the name is young (∼6–10% of pinball in the first trading month). The ablation in Section 2.3 traces the handover from characteristics to the name's own realized volatility as its history lengthens.

7 Conclusion

A fixed-shape parametric filter is hard to beat where its shape assumption holds, and for most assets on most days it holds well. The useful questions are therefore where the exceptions lie and what the flexible model's advantage there consists of. This paper answers the first with an observable signal: a real-time score of residual-shape stress that identifies a state in which a distribution-free family beats the parametric benchmark. It answers the second with a decomposition, which is its central finding. Most of the advantage is the parametric filter's overstatement of variance in the days after an outlier, and a jump-robust GARCH removes that part on its own. A smaller but significant part is conditional-shape information that a robust scale leaves intact. An estimator that exploits both components, while giving up nothing to the benchmark elsewhere, improves on the accuracy of the market-risk models in standard use when judged by one-day diagnostics at the FRTB backtesting levels, at calibration parity with the best of them and short of the desk-level approval a full internal model must pass. Because the estimator pools its residual model across names, it also prices assets with no history of their own, forecasting new listings from their first day as no per-asset model can.

These conclusions come with qualifications. The first is that the score is predictive rather than a necessary or sufficient condition for the flexible model to win. High residual kurtosis coincides with ties in some markets, the nonparametric model wins at only moderate kurtosis in others, and in credit and freight its accuracy edges sit alongside calibration failures that the current tail stages leave open. The second qualification concerns the size of the shape component. The score orders the edge as an empirical regularity, but the decomposition identifies the larger part of that edge as daily-scale misspecification. A GARCH whose news term is capped, so that one shock cannot inflate the variance, removes most of the top-decile edge, and on large capitalizations a five-minute realized-variance scale does the same (Section 5). What survives a robust scale is a conditional-shape component that is real but modest. The evaluation is daily-frequency by design, and the realized-measure comparison qualifies it further. Where intraday data are unavailable or too sparse to build a realized measure, however, the daily flexible model and its score remain the available tool. Finally, the score is a monitor, not a rule for switching models from one day to the next. The day-ahead gap between the oracle and the achievable forecast is largely unforecastable noise (Section 4.3), so the score's value lies in stratifying where the flexible model helps. Distributional uncertainty around these VaR and ES forecasts and the multivariate portfolio tail are left to future work.

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Conflict of interest

The author declares no conflicts of interest.

Ethics statement

This study uses licensed secondary market data (CRSP/WRDS and Bloomberg) under the terms described in the data availability statement. It involves no human subjects and required no ethics approval.

Data and code availability

A public replication package at `github.com/sean-qin-usa/downside-risk-paper` covers every WRDS-based table and figure in this paper and preserves the Bloomberg-based exhibits as run from their derived series. It provides estimation code for all stages of the estimator ((2)–(5)), the misspecification score (9), the FRTB comparison, the generative-posterior and SBC routines, and the walk-forward schedule, seeds, and hyperparameter grids referenced in the text, together with a mapping from each table and figure to the script and result file that produce it. Return data derive from CRSP, Bloomberg, and OptionMetrics under the author's institutional licenses and cannot be redistributed; the package includes the code to rebuild each panel from those sources, the derived non-proprietary series, and a synthetic example dataset on which the full pipeline runs end-to-end without licensed data. The holdout's frozen specification, including its written-in-advance predictions, is recorded in the package alongside the design-era pipeline so that the two can be compared line by line. A citable archival snapshot (Zenodo DOI) will be minted for the accepted version.

Notes

[1]The peaks-over-threshold formulation (Balkema and de Haan, 1974; Pickands, 1975) uses tail data more efficiently than block maxima (Embrechts et al., 1997) and is standard in risk management (McNeil and Frey, 2000). Here $\xi > 0$ is the heavy-tailed case.

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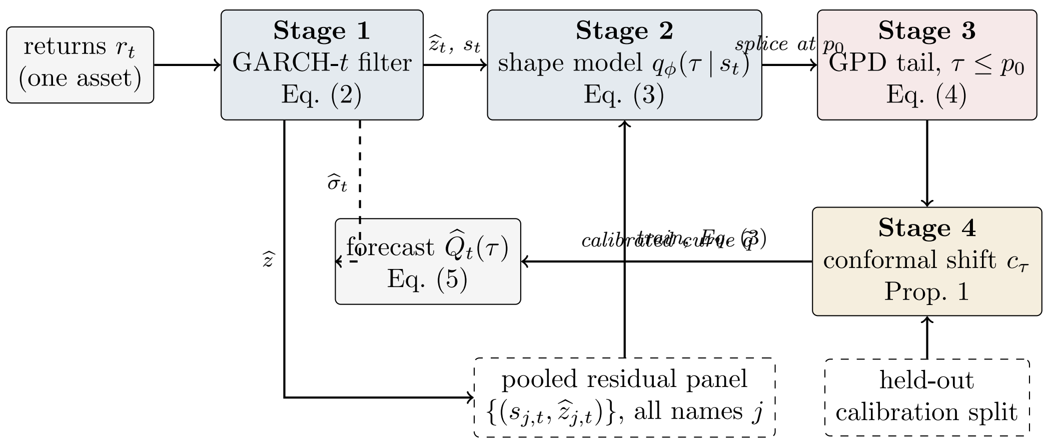

Figure 1: The residual-hybrid estimator as a pipeline. Solid arrows carry data. The dashed top arrow carries the conditional scale $\widehat{\sigma}_t$ that rescales the finished residual quantile curve. Dashed boxes are inputs shared across assets (the pooled panel that trains the amortized shape model) or held out from training (the calibration split behind the conformal overlay). In ongoing use each stage refits annually using only past data. The evaluated protocol freezes each stage once per split, and the annual-refit rerun is reported separately.

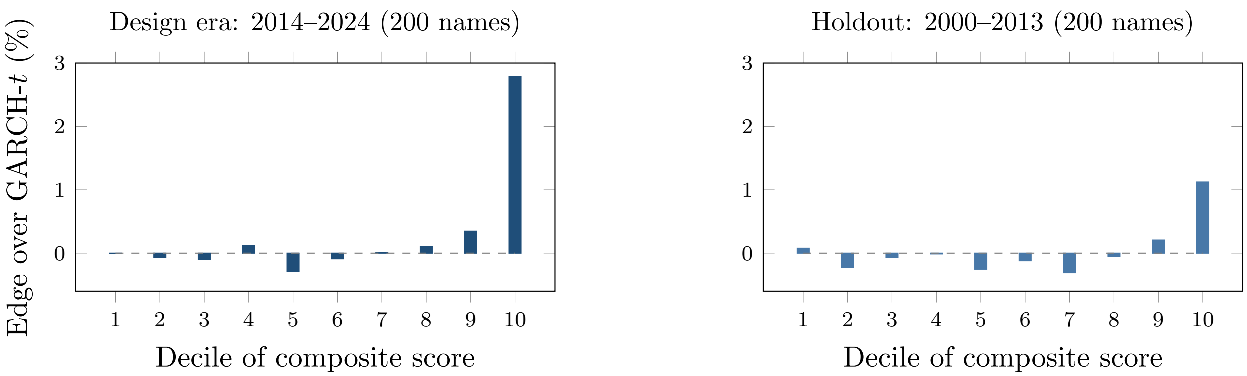

Figure 2: The frontier in and out of era, by decile of the composite misspecification score. Left: the design era (2014–2024). The top decile carries +2.79% at per-date DM 8.9, while deciles one through nine sit within ±0.35%. Right: the composite score run once on 2000–2013 CRSP data. The top decile carries +1.12% at per-date DM 2.51 over 1,465 dates. The threshold shape (edge concentrated in the top decile, near zero elsewhere) survives an untouched era containing the 2008 crisis.

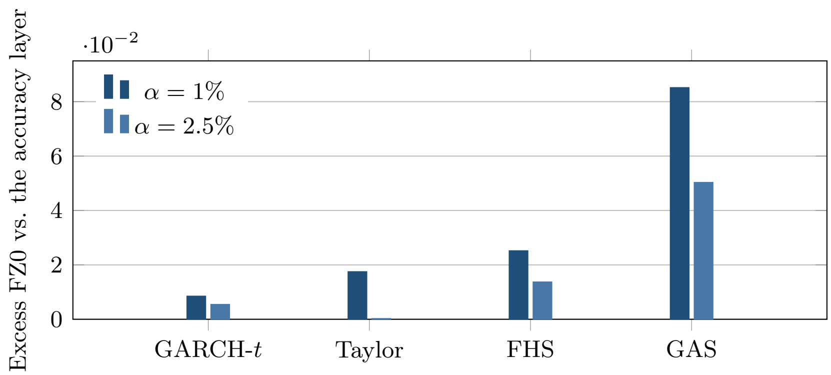

Figure 3: Joint (VaR, ES) FZ0 loss of each benchmark in excess of the residual-hybrid's accuracy layer (the EVT-tailed variant without the conformal shift), on the full 200-name panel. A positive bar means worse than the accuracy layer. The accuracy layer (the zero baseline) attains the lowest FZ0 at both regulatory levels, beating GARCH-t (DM 4.8 at 1%, 5.1 at 2.5%) and FHS (DM 4.9), together with the two FZ-estimated dynamic-ES benchmarks fitted per name by multi-start FZ0 minimization, the GAS of Patton et al. (2019) (DM 9.5 and 8.6) and the ES-CAViaR of Taylor (2019) (DM 3.6 at 1%, a tie at 2.5%).

Table 1: Sample flow across evaluation panels.

Panel	Source	Era	Names	Test obs	Role
Design equity panel	CRSP/WRDS	2014–2024	200	221,600	frontier map, estimator development
Untouched-era holdout	CRSP/WRDS	2000–2013	200	280,608	frozen-spec replication
Exception restatement	CRSP/WRDS	2014–2024	140	≈155,000	per-asset backtesting
Cross-asset sweep	Bloomberg	through 2026	43	[a]	frontier breadth
Country indices	Bloomberg	through 2026	26	[a]	frontier breadth

[a] Frozen exhibits. Per-instrument windows vary and are documented instrument-by-instrument in the repository.

Table 2: The misspecification frontier: nonparametric edge over GARCH-t by decile of the trailing misspecification score (200 CRSP names, 1,108 dates, 221.6k asset-days). Edge = (GARCH pinball − nonparam pinball)/GARCH pinball.

	Edge by signal decile (%)		
Decile	mk_{63} (kurtosis)	$\lvert\text{sk}_{63}\rvert$ (asymmetry)	jump_5
1–8 (bulk)	≈ 0	≈ 0	mixed
9	+0.46	+0.59	—
10 (top)	+**2.98**	+**2.78**	+1.96
Overall		+0.35	

Per-date Diebold–Mariano: top mk_{63} decile +2.98%, DM 10.5; $\lvert\text{sk}_{63}\rvert$ +2.78%, DM 9.85; jump_5 +1.96%, DM 7.63 (all $p \approx 0$). Overall +0.35%, DM 5.35; the composite's bottom decile is indistinguishable from zero (−0.00%, DM 0.36). The composite score's top decile carries +2.79%, DM 8.91 (Figure 2). Top-decile signal means: kurtosis ≈ 25, $\lvert\text{skew}\rvert \approx 4$. jump_5 is U-shaped, its lowest decile also carrying an edge (+1.17%, DM 5.5), so its bulk is mixed rather than near zero.

Table 3: The frontier across universes. Ratio = nonparam/GARCH-t pinball (<1: non-parametric wins). All fits standalone per asset except where noted.

Universe	Segment	Ratio	Resid. kurt.	Significance
FX (24 pairs)	majors (7)	1.019	39	GARCH, pooled DM −13.7
	EM (10)	1.013	8	GARCH, pooled DM −9.5
	crisis (7)	0.973	∼1915	nonparam, pooled DM 4.12
	USDARS	0.880	3298	DM 2.86, $p = .002$
	USDEGP	0.857	1920	DM 3.80, $p = .0001$
Country indices (26)	developed (8)	1.007	4.7	0 nonparam wins
	emerging (10)	1.012	5.3	0 nonparam wins
	frontier (8)	0.993	10.0	4 significant wins[a]
Cross-asset (43)	vol indices	∼0.987	32.8	3/3 significant
	Baltic freight	0.846	—	DM 9.4 (calib. fails)
	IG credit OAS	0.960	—	DM 2.3 (indep. fails)
13/43 significant wins overall, corr(log kurt, edge) = 0.43				
Korea 2026 (23)	all types	1.01–1.12	4–13	GARCH wins all, DM −2.5 to −5

[a] Sri Lanka (DM 3.20), Kenya (3.52), Pakistan (2.38), Nigeria (2.06). Exploratory: only Kenya survives the 93-comparison familywise adjustment (Online Appendix Table OA.4). Cross-index corr(log residual kurtosis, edge) = 0.53. Philippines is the exception (GARCH wins despite kurtosis 13.6), illustrating that kurtosis is neither necessary nor sufficient.

Table 4: FRTB comparison: average pinball loss (twelve quantile levels), $ES_{97.5}$ calibration, and 99% exception tests. 140 CRSP names, 155k test observations.

Model	Avg. pinball	$ES_{97.5}$ pred. vs realized	$Breach_{99}$	$Kupiec_{99}$ p
Ideal value	*(smaller better)*	*(pred. = realized)*	*1.00%*	> 0.05 *(pass)*
Residual-hybrid (GBM)	**0.3551**	−6.37 / −5.83	1.09%	0.00
+ EVT tail	0.3555	−6.92 / −5.91	0.91%	0.00
GARCH(1,1)-t	0.3561	−6.69 / −5.65	1.06%	0.026
GJR-GARCH-skew-t	0.3562	−6.58 / −5.84	1.05%	0.062
GARCH-FHS (per-name)	0.3562	−6.61 / −5.84	1.11%	0.00
GARCH-FHS (pooled)	0.3566	−6.90 / −5.67	0.99%	0.79
GARCH-FHS (rolling 500d)	0.3569	−6.53 / −5.98	1.09%	0.00
EWMA (RiskMetrics)	0.3606	−5.52 / −5.63	1.63%	0.00
Historical simulation	0.3619	−7.21 / −6.86	0.84%	0.00

All columns come from a single common-sample run, provided in the replication package: the 140 CRSP names with the most complete history (at least 1,500 return observations each), over the common 1,108 test dates (155k asset-days), so each per-name exception test is adequately powered. The rolling-FHS burn-in then trims the same early dates from all models identically. Predicted ES is the numerical tail integral $\alpha^{-1}\int_0^\alpha Q(u)\,du$: closed forms for the t and normal entrants, 200-node integration of the Hansen skew-t inverse CDF, and a matched 20-node midpoint rule for the empirical quantile models and for the residual-hybrid, whose VaR and ES are both read from one monotonized min-envelope curve $Q^* = \min\{\text{body,EVT}\}$. A three-node tail-quantile average, a common shortcut, understates predicted ES by 2–6% relative to this integral. Diebold–Mariano vs the residual-hybrid: GARCH-t 5.0, GJR-skew-t 5.0, FHS pooled 6.2, per-name 5.2, rolling 5.7, HS 7.5, EWMA 8.7 (all $p \approx 0$, rounded, and stable across reruns). The EVT variant concedes 0.0004 pinball (DM $\approx$6) to the first row, the raw residual-hybrid without the tail, as the price of that tail. The 90% MCS contains the raw residual-hybrid alone, and CAViaR (separate common-sample run) statistically ties it.

The ES columns are a diagnostic, not a ranking: "realized" is the mean return below each model's own VaR, so the conditioning set is model-dependent, and under the numerical integral each fat-tailed entrant exceeds its own-tail comparator by 9–22%. The joint Fissler–Ziegel score of the FZ0 re-scoring is the ranking criterion for ES.

GJR-skew-t row: Hansen inverse CDF with the fitted η, λ (median 4.2, −0.015), validated against the symmetric-t limit and against an independent reference implementation.

Exception columns are pooled panel diagnostics: at 155k observations a breach-rate deviation of ± 0.1 points rejects in either direction (the EVT variant's 0.91% sits on the conservative side, pooled FHS happens to land at 0.99%), so pooled p-values are reported, not leaned on. The regulatory reading is the per-asset and date-clustered restatement (Online Appendix Figure OA.2): per asset the EVT-tailed residual-hybrid passes Kupiec at 99% for 84% of names and Christoffersen for 86%, and the date-clustered test passes at 99% ($t = 1.01$) and marginally at 97.5% ($t = 1.86$) while the raw residual-hybrid is rejected ($t = 4.5$): the tail stages carry the regulatory calibration, and the 2008-era re-run confirms it out of era. At 97.5% the split-conformal recalibration is the stage that repairs pooled Kupiec (GBM-recal $p = 0.24$).